Dosecheck Rural

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Executive Summary

Problem

Rural volunteer ambulance services carry patients on long transports, often 30 to 70 minutes to the nearest hospital, through areas where cellular coverage is unreliable. Their crews are trained but frequently stretched: many services run with two or three volunteers per call, and the person documenting is usually also the person treating. In interviews with 19 crew members and 6 service directors across four rural counties, 17 of 19 told us they still write vitals and medication doses on paper, gloves, tape or the back of a hand during a call, and reconstruct a formal record afterward. Eleven of the 19 said they had, at least once, been unsure at hand-off whether a repeat dose had been given and when.

The consequences are concrete. Receiving nurses get verbal reports that omit doses or times. Crews are unsure when a second dose is due. Services lose grant eligibility and state reporting compliance when records are incomplete. Existing electronic patient care records exist, but they were designed for paid urban services with reliable connectivity, dedicated tablets and billing departments. Volunteer agencies mostly cannot afford them or cannot use them in a dead zone.

Solution

Dosecheck Rural is an Android application, running on low-cost rugged tablets or crews' own phones, that works fully offline. It does three things.

First, it records vitals and medications with large, glove-friendly buttons. A crew member taps a medication and dose, and the app stamps the time and starts a configurable repeat-dose timer. Vitals entry is designed around a typical sequence of heart rate, blood pressure, oxygen saturation and respiratory rate, with trend lines drawn automatically.

Second, it generates a one-page hand-off report at the hospital door: a vitals trend, a medication log with times and amounts, the crew's notes and the transport timeline. It can be shown on screen, printed on a small portable printer or sent by secure link once a signal returns.

Third, it syncs to a simple agency dashboard when connectivity is available, giving directors complete run records for state reporting and grant applications.

The design principle is that the app must never need a signal and never slow the crew down. Median time to log a medication in our prototype is 6 seconds.

Market

The United States has roughly 15,000 to 20,000 emergency medical services agencies, and volunteer or mostly volunteer agencies make up a large share of them, concentrated in rural areas. For planning purposes we use a conservative figure of 4,000 rural volunteer agencies with an average of 2.5 ambulances each, or about 10,000 ambulances. At our target price this is a serviceable market of roughly 3.5 million dollars annually. This is modest, and we state it plainly: Dosecheck Rural is an initial wedge, not the full opportunity. The same offline hand-off capability applies to rural career services, critical access hospital transfers and, later, other settings with poor connectivity. We have not yet validated demand in those adjacent segments.

Business Model

We sell an annual subscription of 350 dollars per ambulance, with a flat 750 dollars per year for agencies running three or more ambulances. Onboarding is included: we configure medication

lists to each agency's protocols and train crews in one 90-minute session. A hardware bundle with a rugged tablet and portable printer is available at cost plus a small margin for agencies without devices.

Unit economics are favorable on paper. Cloud and support costs are estimated at 45 dollars per ambulance per year, giving about 87 percent gross margin on software. The main uncertainty is customer acquisition. Volunteer agencies have small budgets and long decision cycles, and many fund purchases through annual grants. Our plan is to sell through state and county volunteer EMS associations, which hold regular trainings and can endorse a tool to many agencies at once, and to help agencies write grant applications that cover the subscription. We have not yet closed a paid contract, so the assumption that agencies will pay out of operating budgets or grants remains unproven.

Competition

Three categories compete for the same crews. Paper forms are free and familiar but produce incomplete, illegible records and no dose timing. Full electronic patient care record systems are comprehensive but typically priced for paid agencies, require connectivity or heavy setup and take months to deploy. General-purpose note and form apps are cheap but lack medication timers, hospital-ready reports and protocol-specific lists. Dosecheck Rural does less than a full record system by design: it targets the moment of care and the moment of hand-off, works offline from the first tap and costs a fraction of the alternatives. Our risk is that a large incumbent could add an offline lite mode, though the incumbents we reviewed have shown little interest in the smallest agencies.

Team

Maren Okafor-Lindqvist, chief executive, is a senior in public health who volunteered for three years with a rural ambulance service and led our interview program. Declan Varga, chief technology officer, is a senior in computer science and built the offline sync engine. Priya Thandani, clinical lead, is a nursing student and former emergency medical technician who designed the hand-off report with input from two emergency department nurses. Tobias Reinholt, product and design, is a junior in industrial design who led usability testing with crews. We are advised informally by a retired county EMS director. We lack a team member with regulatory and healthcare compliance experience, which we intend to address through an advisor.

Milestones and Financials

We completed 25 discovery interviews, defined the concept and built a working prototype. We incorporated as an S corporation and launched a website. Two volunteer services are using the prototype free of charge as pilot users, with 112 logged calls to date and a median of 6 seconds to record a medication. Pilot crews rated report usefulness at 4.4 out of 5, and both directors have said they would pay but have not signed commitments.

Our projected first-year plan is 40 ambulances across 16 agencies, producing about 12,000 dollars in subscription revenue against roughly 38,000 dollars in expenses, mainly part-time contractor development, device costs and travel to association trainings. We project break-even at roughly 200 ambulances, which we expect in year three. These figures rest on assumptions about adoption speed that the pilots only partly support. To date we have spent about 4,200 dollars of personal savings.

What We Ask For

We are seeking 25,000 dollars in competition funding. We would use 12,000 dollars to extend development, including a pre-hospital protocol library and an accessibility review; 7,000 dollars to run six paid pilots with state association partners to test willingness to pay; 3,500 dollars for rugged devices and printers; and 2,500 dollars for legal review of data handling and a healthcare compliance consultation. Beyond funding, we ask for introductions to a regulatory advisor with health data experience and to program officers who fund rural emergency services, since securing grant coverage is central to our sales strategy.